

PANCHI TANAJI SHINDE

Quality Engineer

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SUMMARY

- Quality Engineer with 2+ years of experience in statistical process control (SPC), AI-driven predictive maintenance, and quality management systems (QMS) to enhance manufacturing efficiency.
- Skilled in root cause analysis (RCA), failure mode and effects analysis (FMEA), and corrective and preventive actions (CAPA) to improve product quality and compliance.
- Proficient in Python, SQL, Power BI, SAS, Minitab, and SAP QM, with expertise in developing real-time monitoring dashboards and automated quality control systems.
- Strong background in ISO 9001, FDA, ISO 13485, GMP, and Six Sigma methodologies, ensuring strict regulatory adherence and process optimization.
- Led initiatives to reduce non-conformance issues by 30%, implement AI-powered defect detection models, and optimize manufacturing processes for waste reduction and cost efficiency.
- Integrated IoT-enabled sensors and cloud-based analytics (AWS, Azure ML) for real-time condition monitoring, predictive maintenance, and anomaly detection.
- Collaborative problem solver with a proven ability to work cross-functionally, streamline CAPA workflows, and implement Lean Six Sigma (Green Belt) methodologies to drive continuous improvement.

SKILLS

- **Quality Control:** Statistical Process Control (SPC), Root Cause Analysis, Failure Mode and Effects Analysis (FMEA), Corrective and Preventive Actions (CAPA), Inspection and Testing, Quality Audits, GR&R
- **Methodologies:** ISO 9001 Standards, Six Sigma (Green Belt), Lean Manufacturing, Total Quality Management (TQM), Continuous Improvement (Kaizen), Design for Manufacturability, Agile
- **Tools & Software:** Quality Management Systems (QMS), Minitab, SAP, Power BI, Advanced Excel, SQL, SAS
- **Technical Skills:** Data Analysis and Interpretation, Process Mapping and Improvement, Statistical Analysis, Risk Management, Measurement System Analysis (MSA), Process Validation
- **Soft Skills:** Strong Analytical Skills, Attention to Detail, Effective Communication, Team Collaboration, Problem-Solving, Time Management
- **Industry Knowledge:** Manufacturing Processes, Product Lifecycle Management, Supplier Quality Management, Regulatory Compliance (FDA, ISO 13485, ISO 14971, GMP), CpK/PpK Studies
- **Continuous Improvement:** Process Optimization, Waste Reduction, Cost Reduction Strategies, Benchmarking
- **Project Management:** Project Planning and Execution, Resource Allocation, Stakeholder Management, Documentation and Reporting

WORK EXPERIENCE

Honeywell

September 2024 – Present

Quality Engineer

- Implemented machine learning models and IoT sensor data to proactively identify potential equipment failures and reduce downtime.
- Developed a real-time monitoring system using Python, SQL, SAS, and Power BI, enabling predictive analytics that improved maintenance scheduling and reduced unexpected failures by 15%.
- Integrated automated root cause analysis (RCA) and failure mode analysis (FMEA) to enhance asset reliability and optimize maintenance workflows.
- Collaborated with cross-functional teams to implement AI-driven defect detection models, improving production efficiency and reducing maintenance costs.

Dell Technologies

August 2021 – July 2022

Quality Engineer

- Developed a real-time Statistical Process Control (SPC) system to enhance production quality and defect tracking.
- Designed and implemented interactive dashboards using Power BI and SQL, enabling automated data visualization for process deviations and trends.
- Integrated SAP Quality Management Module with SPC dashboards to provide real-time alerts for non-conformance events, reducing defect rates by 25%.
- Collaborated with manufacturing and quality teams to establish control limits, improving response times for corrective actions.
- Conducted statistical data analysis using Minitab to identify process variations and drive continuous improvement initiatives.
- Standardized data collection methods and automated reporting processes, reducing manual inspection efforts and improving efficiency.

- Trained cross-functional teams on the SPC framework, ensuring effective utilization and long-term sustainability of quality monitoring tools.

TatvaSoft **May 2020 – June 2021**

Quality Engineer

- Standardized Corrective and Preventive Actions (CAPA) processes to enhance compliance with ISO 9001 and FDA regulations, reducing non-conformance issues by 30%.
- Designed and implemented a centralized tracking system using SQL, Power BI, SAS and Advanced Excel to monitor CAPA effectiveness and resolution timelines.
- Conducted root cause analysis (RCA) and Failure Mode and Effects Analysis (FMEA) to identify recurring quality issues and develop preventive measures.
- Collaborated with cross-functional teams to streamline CAPA workflows, reducing issue resolution time and enhancing production efficiency.
- Integrated real-time dashboards for tracking CAPA status, enabling proactive decision-making and compliance monitoring.
- Developed and standardized CAPA documentation procedures, ensuring regulatory adherence and improved audit readiness.
- Trained quality assurance teams and manufacturing personnel on CAPA methodologies, improving issue response times and preventive action success rates.

PROJECTS

Smart Manufacturing Quality Control System

- Integrated IoT-enabled sensors with SAP QM for automated real-time defect detection and root cause analysis.
- Designed an SPC dashboard using Tableau and Power BI to monitor key process variables and reduce deviations.

AI-Powered Predictive Maintenance for Manufacturing Equipment

- Developed a predictive maintenance framework using Python, IoT data, and machine learning algorithms to reduce unplanned downtime.
- Integrated cloud analytics (AWS Lambda, Azure ML) for real-time condition monitoring and anomaly detection.

EDUCATION

- **Master of Science in Engineering Management** **August 2022 – December 2024**
California State University Northridge
- **Bachelor of Engineering in Computer Engineering** **May 2016 – June 2021**
Terna Engineering College, Mumbai University

CERTIFICATION

- **Six Sigma Green Belt - California State University Northridge.**
- **Google Project Management Specialization - Coursera.**
- **AI Fundamentals - Data Camp.**